HARSH PANDEY

AI Engineer | Applied AI & LLM Engineer
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PROFESSIONAL SUMMARY

AI Engineer and Computer Science graduate focused on building LLM applications, RAG pipelines, and tool-calling AI agents in Python. Experienced with REST APIs, PostgreSQL, Redis, Docker, and Azure; hackathon winner and TCS CodeVita finalist.

TECHNICAL SKILLS

Programming Languages: Python, SQL
Generative AI & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, Function/Tool Calling, Structured Outputs, Text-to-SQL, Embeddings, Hybrid Search, Reranking, Guardrails
AI Frameworks & Libraries: LangChain, LangGraph, Sentence-Transformers, scikit-learn, Pandas, NumPy, Pydantic, Anthropic Claude API
Vector Databases & Storage: pgvector, PostgreSQL, Redis, SQLite
MLOps & Deployment: FastAPI, REST APIs, Docker, Docker Compose, GitHub Actions (CI/CD), Pytest, Microsoft Azure, Google Cloud

EXPERIENCE

Software Developer Intern Feb 2026 – May 2026
Apisomi *Bhopal, India*

- Engineered 10+ scalable backend REST APIs on a REST architecture for decentralized e-commerce workflows
- Collaborated with 3 cross-functional teams to integrate secure transaction systems, improving platform responsiveness by 25%
- Architected business logic and database operations handling 500+ daily user transactions efficiently
- Optimized SQL queries and API processing, reducing average response time by 30%
- Automated API testing workflows using Postman and Maven, improving debugging efficiency by 40%

PROJECTS

Enterprise RAG Knowledge Assistant | *Python, FastAPI, PostgreSQL + pgvector, Sentence-Transformers, Redis, Claude API, Docker*

- Built a Retrieval-Augmented Generation (RAG) service in Python and FastAPI that answers questions over uploaded PDF and text documents with numbered source citations, backed by PostgreSQL and pgvector
- Implemented hybrid retrieval: HNSW vector search plus Postgres full-text search, merged with Reciprocal Rank Fusion and re-ranked with a cross-encoder
- Grounded the LLM with a citation-required prompt and an "I don't know" fallback when no relevant context is found; added prompt-injection filtering and PII redaction
- Added Redis answer caching with corpus-version invalidation and streaming responses (SSE); containerized with Docker Compose; added unit tests and GitHub Actions CI
- Built an evaluation harness reporting hit@k, MRR, and p50/p95 latency across vector, keyword, hybrid, and re-ranked retrieval

AI Data Analyst Agent | *Python, LangGraph, LangChain, Claude API, FastAPI, Pandas, scikit-learn, SQLite, Docker*

- Built a tool-calling AI agent with LangGraph and LangChain that answers natural-language business questions over a sales database through multi-step planning: schema discovery, SQL generation, statistics, and forecasting
- Designed agent tools for read-only SQL execution with automatic self-correction on errors, pandas summary statistics, and scikit-learn linear-trend forecasting
- Enforced structured outputs with a Pydantic-validated final-answer tool returning the answer, supporting SQL, and a confidence score
- Secured text-to-SQL with two layers: sqlglot validation (single SELECT statements only) and a read-only database connection, plus row and loop limits
- Served through FastAPI and Docker; wrote 29 offline unit tests using a scripted fake LLM and an evaluation harness that scores answers against ground-truth SQL

EDUCATION

Bachelor of Technology in Computer Science and Engineering Nov 2022 – Jun 2026
IES College of Technology, Bhopal | CGPA: 7.44

CERTIFICATIONS & ACHIEVEMENTS

- Earned Microsoft Learn Certifications in Azure API Management, ML Deployment, and Cloud Resource Management
- Achieved Google Cloud Skill Badges in Application Development and Load Balancing on Compute Engine
- Completed Microsoft Learn T-SQL Module focused on database querying and management
- Won Blockchain Hackathon at Amity University among 50+ participating teams
- Secured Finalist position in TCS CodeVita Season 13 (2026) competitive programming contest